I am looking for the info for TEYES (tian zhi yan in chinese pronounce)

I'll check current sources and separate the likely company/brand identity from products, support, and whether it matches the Chinese name you gave.

Thought for 10s >

TEYES is very likely **天之眼** — pronounced roughly **Tiān Zhī Yǎn** in Chinese. It is a Chinese car electronics / car multimedia brand, mainly known for **Android car head units**, navigation systems, DVR/dashcam accessories, reverse cameras, OBD-related accessories, and car infotainment products.

The Chinese company name appears as **深圳市天之眼科技有限公司**, translated roughly as **Shenzhen Teyes Technology Co., Ltd.** A company profile says it was founded in Shenzhen, Guangdong in 2008 and focuses on vehicle navigation R&D, production, and sales. Asian EV

Main product lines I found include:

Product line	Meaning
TEYES CC3 / CC3 2K	Popular Android car stereo/head unit series
TEYES CC4 / CC4 Pro / CC4L	Newer Android car head unit series
LUX ONE	Premium/vehicle-specific head unit series
Accessories	DVR, reverse camera, 360 camera, spare parts, upgrades

The TEYES Australia site lists product categories like **CC4 Pro, CC4L, CC3, CC3 2K, CC3L, LUX ONE, Universal DIN1/DIN2, spare parts**, and support resources like videos, FAQs, tech portal, blogs, reviews, and forum. T'EYES It also shows vehicle-specific support across Toyota, Nissan, Honda, Lexus, Mazda, Volkswagen, Mercedes, BMW, Ford, Jeep, Hyundai, Kia, and others. T'EYES

For your search terms, use these:

```
TEYES
T'EYES
TEYES 天之眼
天之眼车机
天之眼安卓导航
深圳市天之眼科技有限公司
TEYES CC3
TEYES CC4 Pro
```

One important thing: **TEYES is not open-source software** from what I found. It is a commercial hardware/software car head-unit brand. You may find firmware files, ROM discussions, update guides, or community modifications, but that is different from having official open-source code.

Sources

so, is it software?

Not exactly. **TEYES is mainly a hardware + software product brand**, not just software.

Think of it like this:

TEYES = Android-based car multimedia/navigation system

It usually includes:

Part	What it is
Hardware	Car head unit screen, CPU board, RAM/storage, GPS, radio, Bluetooth, amplifier, CANBUS adapter, camera ports
Operating system	Customized Android system for the car screen
TEYES software/apps	Launcher UI, navigation support, DVR/camera app, radio app, Bluetooth app, car settings, CANBUS controls
Firmware	System update files for the head unit

So when people say “TEYES,” they usually mean the **whole car Android head unit**, not only an app.

For example, **TEYES CC3** or **TEYES CC4** is a physical Android car stereo. It runs TEYES customized software on top of Android.

A simple comparison:

```
Phone brand: Samsung
Phone OS/software: Android + Samsung One UI

Car unit brand: TEYES
Car unit OS/software: Android + TEYES car UI/apps
```

So yes, **it has software**, but TEYES itself is better understood as a **car infotainment hardware brand with its own Android software/firmware**.

could you provide me the detailed design info for software and hardware

I’ll break this into two useful layers: what TEYES-like systems usually contain, and what is publicly knowable versus proprietary. I’ll verify recent product/spec info first so we don’t accidentally design around old CC3-era assumptions.

Thought for 4s >

Yes — but one important warning first: **TEYES does not publish full internal schematics, PCB design, firmware source code, or complete software architecture.** So nobody outside TEYES can give the exact official design unless they reverse-engineer the unit. What we *can* describe is the **realistic hardware/software architecture of a TEYES-style Android car head unit**, based on public product info and common Android head-unit design.

TEYES currently sells Android car stereo/head-unit products such as **CC3, CC3 2K, CC4 Pro, CC4L, LUX ONE**, plus accessories such as DVR, cameras, 360 systems, spare parts, and upgrade modules. Their current catalog/support pages show CC4 Pro/CC4L and older

CC3/CC3 2K product families, and their specifications page lists CC3 2K/LUX ONE hardware like Android 10, Unisoc UMS512, octa-core CPU, and 2K display options. T'EYES +1

1. What the product is

A TEYES-like product is basically:

```
Android tablet + car radio + amplifier + GPS + Bluetooth + CANBUS adapter + camera inp
```

It replaces the original car head unit and connects to:

```
Car battery / ACC / ground
Speakers / amplifier
Steering wheel buttons
CANBUS decoder
Reverse camera
DVR / dash camera
GPS antenna
4G SIM / Wi-Fi / Bluetooth
USB
Microphone
Radio antenna
Optional 360 camera module
```

So the design is not just "an app." It is an **embedded Android automotive multimedia system**.

Hardware design

2. Main hardware block diagram

A TEYES-style unit usually looks like this internally:

```
+------------------------------------------------------------+
|                    Android Head Unit                       |
|                                                            |
|  +-------------------+       +------------------------+  |
|  | Main SoC / CPU     |       | RAM + eMMC / UFS Storage|  |
|  | ARM Octa-core      |<----->| 4/6/8GB + 32/64/128GB   |  |
|  +---------+---------+       +------------------------+  |
|            |                                               |
|            | MIPI/RGB/LVDS                                 |
|  +---------v---------+                                    |
```

```
|  | Display + Touch    |  9" / 10" / 12", 2K/QLED etc.     |
|  +-------------------+                                    |
|                                                           |
|  +-------------------+      +-------------------------+  |
|  | Audio DSP / Codec  |----->| Amplifier IC            |  |
|  | EQ, crossover, DAC |      | 4-channel speaker output |  |
|  +-------------------+      +-------------------------+  |
|                                                           |
|  +-------------------+      +-------------------------+  |
|  | MCU Controller     |<---->| CANBUS Decoder / Adapter |  |
|  | ACC, sleep, keys   |      | Car signals             |  |
|  +-------------------+      +-------------------------+  |
|                                                           |
|  +-------------------+      +-------------------------+  |
|  | Radio Tuner        |      | GPS / Wi-Fi / BT / 4G    |  |
|  | FM/AM              |      | Wireless modules        |  |
|  +-------------------+      +-------------------------+  |
|                                                           |
|  +-------------------+      +-------------------------+  |
|  | Camera Inputs      |      | USB / AUX / RCA / HDMI   |  |
|  | Rear/Front/360     |      | External accessories    |  |
|  +-------------------+      +-------------------------+  |
+-----------------------------------------------------------+
```

---

3. Main SoC / processor

Older TEYES CC3 2K/LUX ONE public specs list **Unisoc UMS512**, an **octa-core ARM CPU** using **2× Cortex-A75 + 6× Cortex-A55** cores. T'EYES

For newer CC4 Pro, some reseller/product pages describe newer Android versions, 2K QLED display, and higher-end hardware; one listing mentions Snapdragon-class hardware, but I would treat reseller claims carefully unless confirmed by TEYES official documentation. T'EYES +1

Typical SoC responsibilities:

```
Run Android OS
Render UI
Run navigation apps
Run Bluetooth/CarPlay/Android Auto app layer
Decode video/audio
Handle USB/Wi-Fi/4G
Communicate with MCU
```

```
Communicate with CANBUS app/service
Manage camera preview display
```

Common chip families in Android head units include:

```
Unisoc / Spreadtrum
Qualcomm Snapdragon
Rockchip
MediaTek
Allwinner
UIS7862 / UMS512-style platforms
```

---

4. Memory and storage

Typical configurations:

```
RAM: 2GB / 4GB / 6GB / 8GB
Storage: 32GB / 64GB / 128GB
Storage type: eMMC, sometimes UFS on higher-end models
```

Storage contains:

```
Android system partition
Vendor partition
Boot image
Recovery image
TEYES apps
Maps / user apps
Media files
Logs / cache
Firmware update package area
```

---

5. Display and touch

TEYES product pages describe **2K 2000×1200 QLED** display for CC4 Pro, and TEYES CC3 2K also uses 2K display branding. T'EYES +1

Typical design:

```
LCD/QLED panel
Capacitive touch digitizer
Display interface: MIPI DSI / LVDS / RGB depending on platform
Touch interface: I2C / USB
Brightness control: PWM backlight driver
Auto dimming: MCU or Android receives headlight signal from CAN/illumination wire
```

Important design features:

```
Fast boot / sleep wake
Anti-glare screen
High temperature tolerance
Automotive vibration tolerance
Touch works in heat/cold
Night mode dimming
```

---

6. MCU controller

This is one of the most important parts. In many Android head units, there is a **separate microcontroller unit**, often called the **MCU**.

The Android SoC is powerful, but it is not ideal for low-level car timing/power events. The MCU handles:

```
ACC on/off detection
Ignition state
Sleep/wake control
Power sequencing
Steering wheel key input
Reverse gear trigger
Illumination/headlight signal
Amplifier mute/unmute
CANBUS decoder communication
Button panel communication
Basic watchdog/reset behavior
```

Typical connection:

```
Android SoC <---- UART/I2C/SPI ----> MCU <---- CAN decoder / car harness
```

This is why Android head units have two “firmware” layers:

```
Android firmware
MCU firmware
```

A bad MCU firmware match can break steering wheel buttons, reverse camera trigger, amplifier control, sleep mode, or CANBUS features.

---

7. CANBUS decoder

The CANBUS decoder is usually a small external or semi-external module connected between the car harness and head unit.

It reads vehicle signals such as:

```
Steering wheel controls
Door open/close
Reverse gear
Parking brake
Headlights/illumination
Climate/HVAC info
OEM amplifier state
Vehicle speed
Radar/parking sensor info
Air conditioner panel info
```

Then it translates those vehicle-specific CAN messages into a simpler protocol for the head unit.

Typical architecture:

```
Car CAN High/Low
      |
      v
CANBUS Decoder Box
      |
      v
MCU / Android service
      |
      v
TEYES car settings UI
```

This is the reason head units are often **vehicle-specific**. A Toyota harness/CAN profile is different from BMW, Nissan, Hyundai, etc. TEYES Australia lists vehicle support across many brands, including Toyota, Nissan, Honda, Lexus, Mazda, Volkswagen, Mercedes, BMW, Ford, Jeep, Hyundai, Kia, and others. T'EYES

---

8. Audio subsystem

A car head unit needs better audio hardware than a normal tablet.

Typical audio path:

```
Android App / Media Player
        |
        v
Android AudioFlinger / HAL
        |
        v
Audio Codec / DSP
        |
        v
Power Amplifier IC
        |
        v
Car Speakers
```

Possible hardware blocks:

```
Audio codec chip
DSP chip
Power amplifier chip
Pre-out RCA outputs
Subwoofer output
Optical/coaxial output on some models
Microphone input
Radio tuner audio input
Bluetooth audio path
```

Common software audio features:

```
Equalizer
Time alignment
Crossover
Subwoofer level
Bass enhancement
Loudness
Balance/fader
```

```
Per-source volume
Navigation voice mixing
Reverse mute
Bluetooth call echo cancellation
```

Some TEYES listings mention DSP/built-in equalizer, 5.1 output, and coaxial/fiber optic outputs depending on model/listing. eBay +1

---

9. Radio tuner

Radio is normally not handled by Android directly. It is usually handled by a tuner IC.

Hardware:

```
FM/AM antenna input
Radio tuner IC
Audio output to codec/DSP
Control via I2C/SPI/UART
```

Software:

```
Radio app
Station scan
RDS if supported
Favorite stations
Region settings
```

---

10. Connectivity hardware

Typical modules:

```
Wi-Fi module
Bluetooth module
GPS/BeiDou/GLONASS receiver
4G LTE modem / SIM slot
USB controller/hub
MicroSD slot on some models
External microphone
```

The product listings commonly mention Wi-Fi, Bluetooth, GPS, Apple CarPlay, Android Auto, USB input, rear camera input, steering wheel control, and 4G SIM support. eBay

11. Camera subsystem

Common inputs:

```
Rear camera input
Front camera input
DVR dashcam USB input
360 camera module input
AHD/CVBS analog camera input
```

Reverse camera flow:

```
Driver selects reverse gear
        |
CANBUS decoder or reverse trigger wire detects reverse
        |
MCU wakes camera input path
        |
Android UI switches to reverse camera app/activity
        |
Parking lines / radar overlay displayed
```

360 camera flow:

```
4 cameras
   |
360 camera processor/module
   |
Processed video output to head unit
   |
Android displays around-view image
```

TEYES support pages include how-to content for CC4 Pro 360 system activation and hardware topics like changing 2K screens, which confirms that camera/display hardware servicing is part of their ecosystem. T'EYES

---

12. Power design

Automotive power is harsh. A real head unit cannot just use a simple 12V adapter.

Power inputs:

```
Battery 12V constant
ACC ignition 12V
Ground
Illumination
Reverse trigger
Parking brake
CAN signals
```

Power design needs:

```
12V to 5V buck converter
5V to 3.3V / 1.8V / 1.2V regulators
Load dump protection
Reverse polarity protection
Fuse
EMI filtering
Sleep current control
Thermal protection
Brownout handling during engine crank
```

Sleep/wake behavior:

```
ACC off:
  Android goes to sleep or shuts down after delay
  MCU remains low-power awake

ACC on:
  MCU wakes SoC
  Android resumes quickly
```

That “fast boot” experience is usually not a real cold boot; it is often suspend/resume controlled by the MCU.

Software design

13. Software stack overview

A TEYES-like software architecture looks like this:

```
+------------------------------------------------------+
|                  User Applications                   |
| Navigation, Music, Radio, DVR, Camera, Settings      |
| CarPlay/Android Auto, Bluetooth Phone, TEYES UI      |
```

```
+----------------------------------------------------+
|                 TEYES System Apps                  |
| Launcher, Car Settings, CANBUS UI, Audio DSP UI    |
| Update App, Vehicle Config, 360 Camera App         |
+----------------------------------------------------+
|                 Android Framework                  |
| Activity Manager, Audio, Bluetooth, Location, USB  |
+----------------------------------------------------+
|                 Vendor Services / HAL              |
| MCU service, CAN service, Radio HAL, Audio HAL     |
| Camera HAL, GPS HAL, Power service                 |
+----------------------------------------------------+
|                 Linux Kernel + Drivers             |
| Display, Touch, USB, Wi-Fi, BT, GPS, Audio, UART   |
+----------------------------------------------------+
|                 Bootloader / Recovery / MCU FW     |
+----------------------------------------------------+
```

14. Android OS layer

Public TEYES pages mention different Android versions depending on product family. CC3 2K/LUX ONE specs show Android 10, while CC4 Pro product text mentions TEYES OS based on Android 13 on one official-region page. T'EYES +1

Typical Android layers:

```
Bootloader
Linux kernel
Android system image
Vendor image
HAL drivers
System services
TEYES apps
Google apps / Play Store depending on region
```

Key customizations:

```
Custom launcher for car UI
Large touch buttons
Split screen support
Boot animation
Car settings panel
DSP control
CANBUS profile selection
```

```
Reverse camera handling
Steering wheel key mapping
Firmware update app
```

---

15. Main software modules

A. Launcher / home UI

Responsibilities:

```
Home screen
Navigation shortcut
Music widget
Bluetooth phone widget
Radio widget
Vehicle status widgets
App drawer
Theme / wallpaper
Split-screen layout
```

B. Vehicle settings app

Responsibilities:

```
CANBUS model selection
Car type selection
Steering wheel key learning
Reverse camera settings
Door/radar/HVAC display settings
Amplifier settings
Boot logo selection
Factory settings / hidden settings
```

C. MCU communication service

This is a background Android service talking to MCU.

Example responsibilities:

```
Read ACC state
Read reverse state
Read steering wheel buttons
Read CANBUS events
```

```
Send volume/mute commands
Send sleep/wake command
Send backlight brightness
Receive watchdog/health messages
```

Example internal protocol idea:

```json
{
  "type": "STEERING_KEY",
  "key": "VOLUME_UP",
  "pressed": true,
```

JSON

Or binary UART frames like:

```
AA 55 04 10 01 02 CS
```

The real TEYES protocol is proprietary, but the concept is common.

D. CANBUS service

Responsibilities:

```
Decode vehicle-specific signals
Normalize events into common format
Expose events to apps
Update UI overlays
Handle vehicle-specific quirks
```

Normalized event examples:

```json
{
  "vehicle_event": "DOOR_STATUS",
  "front_left": "OPEN",
  "front_right": "CLOSED",
  "rear_left": "CLOSED",
```

JSON

```json
{
  "vehicle_event": "HVAC_STATUS",
  "temperature_left": 22.5,
  "temperature_right": 23.0,
  "fan_level": 3,
```


JSON

E. Radio app/service

Responsibilities:

```
Tune FM/AM
Scan stations
Save presets
Region selection
RDS data
Audio source switching
```

F. Audio/DSP app

Responsibilities:

```
EQ bands
Loudness
Subwoofer
Crossover
Time alignment
Balance/fader
Source gain
Factory audio profile
```

G. Bluetooth phone/media

Responsibilities:

```
Pair phone
Hands-free calling
Contacts sync
A2DP music
Microphone routing
Echo cancellation
Call UI
```

H. CarPlay / Android Auto integration

Many TEYES listings and reviews mention Apple CarPlay and Android Auto support.

eBay +1

Possible architecture:

```
Phone connects by USB or wireless
        |
CarPlay/Android Auto projection service
        |
TEYES projection app renders phone UI
        |
Audio routed through Android audio system
        |
Touch input forwarded back to phone
```

I. Camera/reverse/360 app

Responsibilities:

```
Show rear camera automatically
Overlay parking guide lines
Show radar/PDC information
Switch front/rear/left/right cameras
Handle 360 module activation
Record DVR video if supported
```

J. OTA / firmware update app

Responsibilities:

```
Check current Android version
Check MCU version
Install update package from USB/local/online
Reboot into recovery
Verify package signature/checksum
```

```
Update system/vendor/boot/recovery partitions
Update MCU firmware separately
```

---

16. Data flow examples

Reverse camera flow

```
Reverse gear selected
      |
CANBUS decoder or reverse wire detects signal
      |
MCU sends REVERSE_ON event
      |
Android MCU service receives event
      |
Camera app opens immediately
      |
Audio may mute/reduce
      |
Parking lines/radar overlay shown
```

Steering wheel button flow

```
User presses steering wheel volume +
      |
Car CANBUS message generated
      |
CANBUS decoder translates message
      |
MCU receives key event
      |
Android MCU service receives VOLUME_UP
      |
Android audio manager increases volume
```

Radio audio flow

```
Radio tuner IC receives FM
      |
Analog/digital audio goes to codec/DSP
      |
Android radio app controls station frequency
```

```
      |
DSP/amplifier outputs to car speakers
```

Navigation voice mixing

```
Music playing
      |
Navigation app gives voice instruction
      |
Android audio focus lowers music volume
      |
Navigation voice mixed through speaker
      |
Music returns to normal volume
```

---

17. If you wanted to build a TEYES-like product

For your own design, the architecture should be:

Hardware BOM-level modules

```
1. Main board
   - ARM SoC
   - RAM
   - eMMC/UFS
   - PMIC
   - Wi-Fi/BT
   - GPS
   - LTE module optional

2. Display board
   - LCD/QLED panel
   - Touch controller
   - Backlight driver

3. Audio board or audio section
   - Audio codec
   - DSP
   - Amplifier IC
   - RCA pre-outs
   - Microphone input

4. Vehicle I/O board
   - MCU
```

```
   - CAN transceiver
   - UART/I2C/SPI bridge
   - Reverse/illumination/parking brake inputs
   - Steering wheel key input

5. Power board
   - Automotive 12V protection
   - DC/DC converters
   - Sleep current control
   - Fuse/filtering

6. Camera/video section
   - CVBS/AHD decoder
   - USB camera support
   - 360 camera module interface

7. Harness/accessories
   - Vehicle-specific connector
   - CANBUS box
   - GPS antenna
   - 4G antenna
   - USB cable
   - RCA cable
```

Software modules

```
1. Android base OS
2. Vendor HAL/drivers
3. MCU communication daemon
4. CANBUS abstraction service
5. Audio/DSP service
6. Radio service
7. Camera/reverse service
8. Bluetooth/phone service
9. Launcher/UI
10. Settings/factory app
11. OTA/update system
12. Diagnostics/logging
```

Recommended service architecture

```
mcu-service
```



```
canbus-service
  normalizes car-specific signals
```

```
vehicle-state-service
  keeps latest car state: reverse, door, speed, lights, HVAC

audio-service
  controls DSP, volume, source, mixing

camera-service
  handles reverse/front/360 camera switching

radio-service
  controls tuner IC

settings-service
  stores car profile, key mapping, user preferences

update-service
  firmware and MCU update workflow

launcher-app
  user-facing home screen

factory-settings-app
  advanced hidden configuration
```

A clean version would separate it like this:

```
Hardware drivers  ->  Vendor services  ->  Android apps/UI
```

Do **not** put CANBUS decoding directly inside the launcher. That becomes a spaghetti monster with wheels.

---

18. Database / storage design inside the head unit

Local storage can use:

```
SQLite
SharedPreferences
Android Room DB
Flat config files
```

Example tables/configs:

```
vehicle_profile
- brand
- model
- year
- canbus_protocol
- amplifier_type
- camera_type

steering_key_mapping
- physical_key
- android_action

audio_profile
- eq_band_values
- subwoofer_level
- balance
- fader
- time_alignment

system_settings
- boot_logo
- theme
- brightness
- reverse_line_enabled

device_info
- android_version
- mcu_version
- serial_number
- activation_status
```

---

19. Important engineering challenges

The hard parts are not the Android UI. The hard parts are:

```
Stable power/sleep/wake behavior
CANBUS compatibility per vehicle
Audio noise/ground loop control
Fast reverse camera switching
Thermal stability
Firmware update safety
EMI/EMC compliance
Touch/display durability
Boot speed
```

```
CarPlay/Android Auto licensing/integration
Factory amplifier compatibility
```

In car electronics, one “small” issue can make the product feel bad:

```
Battery drain overnight
Reverse camera opens 2 seconds late
Steering wheel buttons mismatch
Radio has weak reception
Bluetooth call has echo
Amplifier pops when ACC turns on
Screen too dim in sunlight
CANBUS profile breaks AC display
```

That is the swamp. Bring boots.

---

20. What is probably proprietary in TEYES

These parts are almost certainly closed/proprietary:

```
PCB schematic
MCU firmware
CANBUS decoder firmware
TEYES launcher source code
DSP control app source code
Factory settings app
OTA update package signing
CarPlay/Android Auto integration layer
Vehicle profile database
Activation/licensing system
360 camera calibration logic
```

What you may find publicly:

```
User manuals
Installation videos
Firmware update files
Community reverse-engineering notes
XDA/forum discussions
Product specs
Wiring diagrams for some vehicles
```

But that is not the same as full source code or official hardware design.

Bottom line

TEYES is best understood as an **Android automotive embedded system**. Its design combines:

```
Automotive hardware
Android OS
MCU firmware
CANBUS vehicle integration
Audio DSP/amplifier system
Camera/radio/Bluetooth/GPS modules
Custom TEYES apps and launcher
```

If your goal is to **build a similar product**, start with this architecture:

```
ARM SoC board + Android BSP
MCU board for vehicle/power control
CANBUS decoder layer
Audio DSP + amplifier
Display/touch module
Camera input module
Custom Android launcher + vehicle services
Vehicle-specific harnesses
```

For a first prototype, I would not design the whole PCB from zero. I'd use an existing Android head-unit development board or automotive SoM, then build your own **MCU + CANBUS + launcher software** around it. That gives you a working prototype without drowning immediately in PCB, EMI, bootloader, and Android BSP problems.

Sources

could you provide me some pdf which is related to the technical info for the product?

I'll look for official TEYES manuals/spec PDFs first, then add closely related technical PDFs for Android head-unit hardware/software architecture if TEYES doesn't publish deep engineering docs.

Thought for 20s >

Yes. Here are the best PDF / technical-document sources I found.

TEYES-specific docs

1. **TEYES CC3 2K User Manual PDF**
   This is the most useful TEYES-related PDF I found. It includes technical specs: Android 10, UIS7862S 8-core CPU, 2K 2000×1200 display, AHD rear camera input, Wi-Fi, 4G LTE, Bluetooth chip, TDA7851 audio amp, ADAU1701 DSP, GPS/BeiDou/GLONASS, etc. Use the **Download PDF** button on the page. Manuals+
2. **TEYES official Technical Portal**
   Not always PDF, but useful for product technical support: firmware, diagrams, CANBUS settings, OEM pinouts, apps, faults/solutions, how-to guides. T'EYES
3. **TEYES CC3 Wiring Diagram**
   Good for understanding external hardware connections: harness, camera, RCA, USB, CANBUS, power, speaker wiring. The official TEYES menu also links brochures and comparison charts for CC4 Pro, CC4L, LUX ONE, CC3 2K, CC3, CC3L, etc. T'EYES
4. **TEYES CC3 CANBUS Settings**
   Useful for understanding what the CANBUS adapter controls: steering wheel buttons, reverse display, door info, original-car functions, etc. T'EYES
5. **TEYES CC4 Pro Brochure / CC3 2K Information / Specification Comparison**
   These are official product-info pages rather than deep engineering PDFs, but they help compare models and hardware generations. T'EYES +2

Component-level technical PDFs related to TEYES-style hardware

These are not TEYES manuals, but they are useful because TEYES CC3 2K documentation mentions similar chips/modules.

PDF	Why useful
STMicroelectronics TDA7851 / TDA7851A datasheet	Car-radio 4-channel MOSFET power amplifier, useful for speaker output design. STMi... +1
Analog Devices ADAU1701 datasheet	Audio DSP used for EQ, crossover, delay, bass enhancement, audio processing. Analog D...
Analog Devices ADAU1701 Mini Evaluation Board PDF	Better than only the datasheet if you want reference hardware wiring/design around the DSP. Analog Devices
Realtek RTL8761 Bluetooth datasheet	Related to Bluetooth controller design; the TEYES manual mentions Realtek 8761 Bluetooth. Make Elec...

What you probably will *not* find publicly

TEYES likely does **not** publish these:

```
Full PCB schematic
Board layout files
MCU firmware source code
Android BSP source code
CANBUS decoder firmware
TEYES launcher source code
OTA signing/update system design
Factory settings app source code
```

For practical technical study, start with this order:

```
1. TEYES CC3 2K User Manual PDF
2. TEYES CC3 Wiring Diagram
3. TEYES CANBUS Settings
4. TDA7851 datasheet
5. ADAU1701 datasheet + evaluation board PDF
6. Realtek RTL8761 datasheet
```

That gives you the best public picture of the product's hardware/software design without needing TEYES internal documents.

Sources